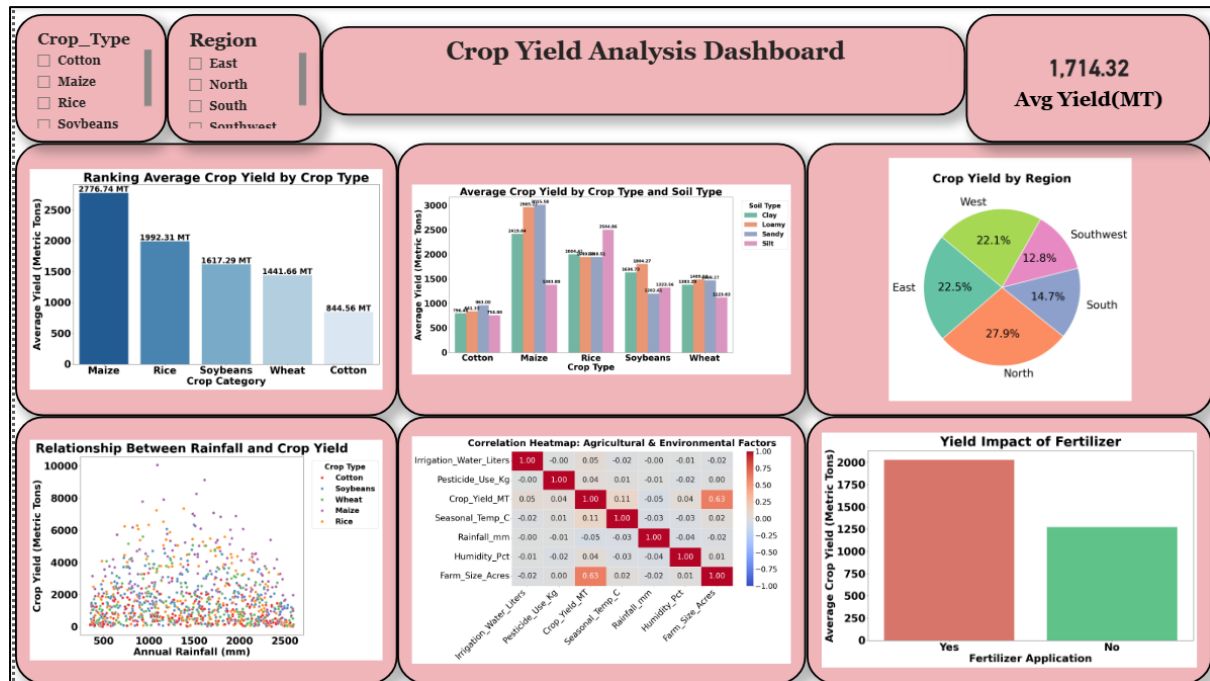


Crop Yield Analysis and Data Cleaning using Python

Dashboard:



Insights Gathered:

- **Crop Production:** Maize is the dominant volume contributor with a peak average yield of 2,776.74 MT, whereas Cotton registers the lowest footprint at 844.56 MT.
- **Regional Contribution:** Total agricultural production is highly concentrated in three zones: the North (27.9%), followed closely by the East (22.5%) and West (22.1%), while the Southwest lags significantly behind as the lowest producer at 12.8%.
- **Resource Impact:** Farm plots treated with fertilizer achieve a substantially higher average output of ~2,025 MT, compared to unfertilized plots which stop at ~1,275 MT (a clear 750 MT production gap).
- **Scale Driven Growth:** The strongest mathematical driver of harvest volume is physical land scale, proved by a strong positive correlation of 0.63 between Farm_Size_Acres and Crop_Yield_MT.
- **Environmental & Soil Preferences:** Specific crop yields are highly sensitive to their soil mediums. Soybeans (1,804.27 MT) and

Wheat (**1,489.16 MT**) thrive optimally in Loamy soil; Maize (**3,015.58 MT**) and Cotton (**963.00 MT**) adapt best to Sandy soil; while Rice (**2,504.86 MT**) maximizes its performance strictly in Silt soil.

- **The Rainfall Sweet-Spot:** Yields do not scale infinitely with water. There is an optimal climate zone where crop yields peak dynamically between **1,000 mm and 1,600 mm** of annual rainfall. Beyond this window, extreme drought or flooding conditions act as stressors that suppress overall growth.

Recommendations:

- **Implement Precise Soil-to-Crop Mapping:** Mandate regional crop distribution strategies that align crops with their top-performing soil environments (e.g., dedicate Sandy soils entirely to Maize/Cotton and Silt basins exclusively to Rice).
- **Scale Fertilizer Deployment:** Prioritize fertilizer distribution frameworks to unfertilized or underperforming farm plots, as this single management practice guarantees an average yield lift of roughly **750 MT** per plot.
- **Target Regional Infrastructure in the Southwest:** Investigate the structural bottlenecks causing the Southwest region to produce a mere **12.8%** of total output. Deploy irrigation systems or land consolidation efforts there to build resilience against its lower yields.
- **Develop Climate-Resilient Water Management:** For regions experiencing rainfall outside the optimal **1,000 mm – 1,600 mm** boundary, install smart drainage systems to handle flooding and water-retention basins/reservoirs to safeguard against droughts.

Conclusion:

From the analysis, we identified meaningful relationships and patterns in the agricultural dataset. The visualizations and multi-variable insights highlight that physical land scale, strategic fertilizer deployment, the optimal rainfall spot and precise crop-to-soil mapping are the dominant factors influencing crop yields and this analysis provides a strong, data-driven foundation for maximizing future harvest volumes and making smarter, evidence-based farming decisions.